

CSE 370– Database Systems

Assignment 3

Spring 2025

Submission Instructions:

1. Write your name, id, section on top of the first page
2. Your answer should be handwritten, take pictures and create a single pdf.
3. **Submit the pdf in the following form:** <https://forms.gle/aEQiP5b2BdSoG59o9>
4. **Submission Deadline:** 15th May 2025 (Thursday), 11:59 pm (midnight)
5. **NO LATE SUBMISSION WILL BE ACCEPTED. Start Early!**

Question 1 [10 Marks]

- A. **Construct** a B+ tree of order $n = 3$ for the following search key values inserted in the given order: Apple, Orange, Grapes, Kiwi, Banana, Pineapple, Pear, Watermelon, Blueberry. Show the expansion of the tree for each insertion and the final tree. Values in nodes are sorted in alphabetical order.
- B. **Delete** the following from the B+ tree constructed in (A): Banana, Pear, Blueberry, Grape, Kiwi, Orange.

Question 2 [10 Marks]

Suppose you have the table below for “Instructors” for a School Database with columns ID, Name, Subject and Salary respectively:

76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
45565	Katz	Comp. Sci.	75000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000
12121	Wu	Finance	90000
76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
33465	Gold	Physics	87000

Construct a hash index on attribute ID. Suppose the number of buckets in the hash index is 7 and each bucket can house 2 index entries at max. Bucket overflow is handled using forward chaining. The hash function, $h = (\text{first digit} + \text{middle digit} + \text{last digit in ID}) \% 7$.

E.g. $h(76766) = (7 + 7 + 6) \% 7$

Question 3 [10 Marks]

Sakib and Sarah are trying to reserve the same book- “DB Fundamentals” from an online library system. 5 copies of the book are remaining. T1 and T2 represent Sakib’s and Sarah’s transactions respectively. The concurrent transactions happen in the sequence given below:

1. T1: Sakib saw 5 copies left and reserved 1 book for himself.
 2. T1: The number of copies is updated to 4, but now committed.
 3. T2: Sarah also saw 5 copies left and reserved 2 copies for herself and her friend.
 4. T2: The number of copies of the book is updated to 3 and is committed.
 5. T1: The number of copies is updated to 4 from Sakib’s transaction.
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- a. Explain which ACID properties have been violated by the above transaction.
 - b. State and explain all concurrency issues that have occurred due to the above transaction.
 - c. Explain solutions for avoiding the concurrency issue mentioned in (b).